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Growth of *Acropora pulchra*

II. Responses of natural and transplanted colonies to temperature and day length *

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Abstract

The growth of both undisturbed colonies and transplants of the branching coral *Acropora pulchra* (Brook) was studied over a 17-month period in the reef of Bolinao, Pangasinan, northern Philippines. Growth rate was monitored using linear measurement techniques. Of the environmental factors measured, both temperature and day length, in the range of natural conditions encountered, correlated negatively with the growth and branching of the undisturbed corals and, to a lesser extent, growth of the transplants. Temperature, in addition, appeared to affect mortality of the transplants significantly. Using various methods of computation of data for the undisturbed colonies, the projected annual growth rate of the species was found to range from 13.1 to 15.8 cm yr^{-1}. Growth rates of the transplants were in general considerably lower.

Introduction

In a previous paper (Yap and Gomez, 1981), the growth rate of the branching coral species *Acropora pulchra* was shown to be negatively correlated with temperature. Mortality likewise appeared to be significantly related to temperature, the latter's effects possibly compounded by additional stress caused by sedimentation. The present study covers the period October 1980 to March 1982, building on and concluding the first study, which spanned the months of October 1980 to May 1981. It includes analyses of records of an additional factor, day length, here representing an alternative means of measuring light. Day length, along with temperature, is shown to correlate negatively with coral growth rate. These factors together characterize the warm summer months believed to be stressful for growth of the species in shallow water.

Observed trends in the initiation of branching ("budding") throughout the year are related to trends in growth rate. Data on the regeneration of broken branches within a year's time are also presented, and related to projected annual growth rates of the species.

Relatively few studies exist on the effects of temperature, acting alone or in concert with other factors, on coral growth in the tropics and subtropics (Clausen, 1971; Jones and Randall, 1973; Jokiel and Coles, 1974, 1977; Coles *et al.,* 1976; Hudson, 1981). In some cases, sublethal elevation of temperature has been shown to reduce growth rates in certain coral species (Jones and Randall, 1973). High light intensities combined with extreme temperatures likewise appeared to affect coral physiology adversely (Coles and Jokiel, 1978).

Material and methods

Location

Bolinao, Pangasinan is located at the northwestern corner of Lingayen Gulf, northern Philippines, approximately between Lat. 16°21′ and 16°28′N, and between Long. 119°53′ and 120°00′E. The study area, situated in the lagoon of Santiago Island in Bolinao, is described in detail in Yap and Gomez (1981). Sites A, B and D are natural *Acropora pulchra* (Brook) patches, while Site C is a rubber-tire artificial reef divided into four groups of tires called "modules" and situated in a bare, silty portion of the reef. Depths vary from 3 to 5 m, depending on the tide.

Measurement of growth and transplantation of corals

At least 15 naturally-established colonies growing side by side in each of Sites A, B and D were designated as the controls. Two branches per colony (to make a total of at least 30 branches per site) were tied with plastic-coated copper wire. This served as a means of identification as

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well as a measuring base. Linear branch increment was monitored by measuring branch length from the wire base to the tip at monthly intervals with an architect's plastic flexible curve graduated in millimeters.

Other colonies at Sites A and B served as sources of the transplants. A minimum of 15 coral fragments were broken off and reattached within their same sites of origin (A and B), as well as transplanted to Site C, the artificial reef. A corals were transplanted to Module IV, and B corals to Module I. Transplants were attached to rubber tires and concrete flagstones by means of underwater epoxy. Two branches per transplant were tagged and monitored in the same manner as the controls.

Mortality was taken to be the number of dead, marked branches over the total number of marked branches at each site.

Measurement of environmental parameters

Records of day length were procured from weather reports. Temperature was continuously monitored with a Peabody-Ryan Model "J" Thermograph tied to the bottom of the artificial reef.

Results

Environmental factors

Mean values for water temperature and day length for each measurement interval within the observation period spanning October 1980 to September 1981 are plotted in Fig. 1 a. Shorter days were generally associated with lower water temperatures, although the extreme lows of each did not coincide. The shortest average day length, 11.29 h, was recorded for the period December 1980 to January 1981, while mean temperature was lowest at 26.0 °C for the period January to February 1981. No data are available for later months.

Growth

Trends in growth of controls and transplants were inferred from the data by means of analysis of variance (ANOVA) tests (Sokal and Rohlf, 1969).

Controls

Mean monthly growth rates of the control colonies at each site are depicted in Fig. 1 b, c, and d. These ranged from a low of 0.3 to a high of 2.3 cm/mo. Negative increments, such as those brought about by mechanical breakage or predation, were excluded from the averages. In general, no significant differences in growth rates among the different sites were registered.

Significant trends were revealed upon testing for variation of growth rates over time. Monitoring of growth started well into the cool season (October 1980) when growth rates were at their high points. With the onset of the warmer months (April to June), growth rates at all

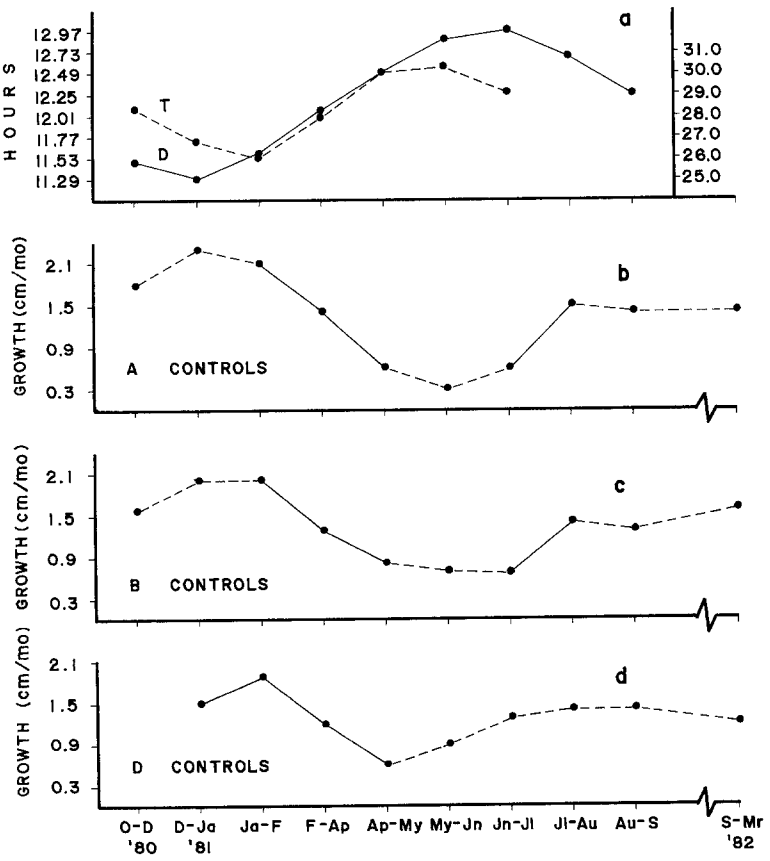

Fig. 1. *Acropora pulchra.* (a) Temperature and day length curves for the months October 1980–July 1981 and October 1980–September 1981, respectively. T= Temperature, D=Day length. (b–d) Mean monthly growth rates (cm/mo) of control colonies at Sites A, B, and D, from October 1980 to March 1982. Solid lines represent significant differences ($P=0.05$) between data points, and dashed lines insignificant differences. Data for Site D from December 1980

sites decreased significantly, and remained at a constant low for the duration of the entire summer. The onset of the rainy months of July and August was again accompanied by a significant growth spurt of the corals. Data for the interval September 1981 to March 1982 represent increments over a six-month period and are similar to growth rates observed for the cool months.

Transplants

Trends in mean monthly growth rates of the transplants are presented in Fig. 2. Values ranged from 0 to 1.1 cm/mo. As in the case of the controls, no significant differences existed among transplants at the different sites. A pattern could also be discerned of higher growth rates during the cool season.

As a corollary to the monitoring of growth rates, the percentages of branches of transplants that did not register any growth for a particular time interval were recorded for each site. The greater percentages of branches registering zero growth rates were obtained during the warm season roughly spanning the months of April to June (Fig. 3).

Regeneration of broken branches

Data on the regeneration and growth of broken branches are presented in Table 1. Percentage mortality (i.e. of branches that did not survive the entire monitoring period) was 36.4%. This figure does not include branches that could not be located at the end of the observation period. These preliminary observations indicate that an average of 12.6 cm of new growth may be regenerated in a year's time, provided the coral survives the trauma of breakage and concomitant stresses.

Mortality (transplants)

Percentage mortalities of branches of transplants are plotted in Fig. 4. Comparisons of responses of transplants among the different sites showed no significant differences. On the other hand, increases in percentage mortality for transplants at all sites again occurred during the warm months.

Budding

Data on the initiation of new branches on the same observed branches at Sites A, B and D for each time interval within the study period are presented in Table 2. Branches that did not bud during the entire observation period were not included in the computations. These "dormant" branches comprised 3.7, 26.5 and 14.3% of the total branches monitored at Sites A, B and D respectively. As indicated by the data in Table 2, budding seems to have been more profuse during the cooler season as compared to the warmer season – the same period when growth rates were also relatively high.

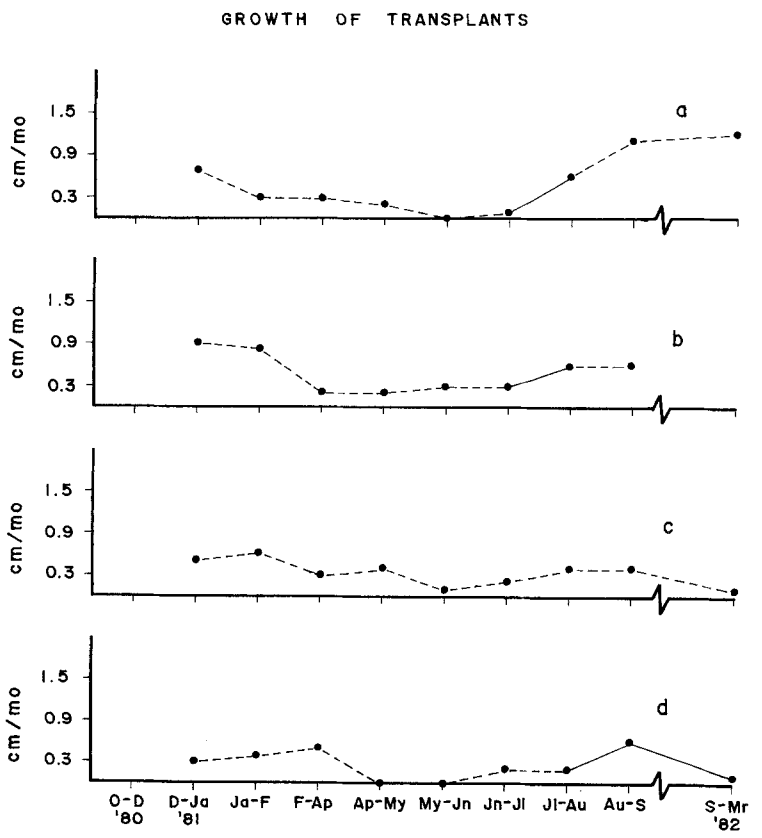

Fig. 2. *Acropora pulchra.* Mean monthly growth rates (cm/mo) of transplants from December 1980 to March 1982. Solid lines represent significant differences ($P = 0.05$) between data points, and dashed lines insignificant differences. (a) Transplants at Site A; (b) Transplants at Site B; (c) Transplants at Module I, Site C; (d) Transplants at Module IV, Site C

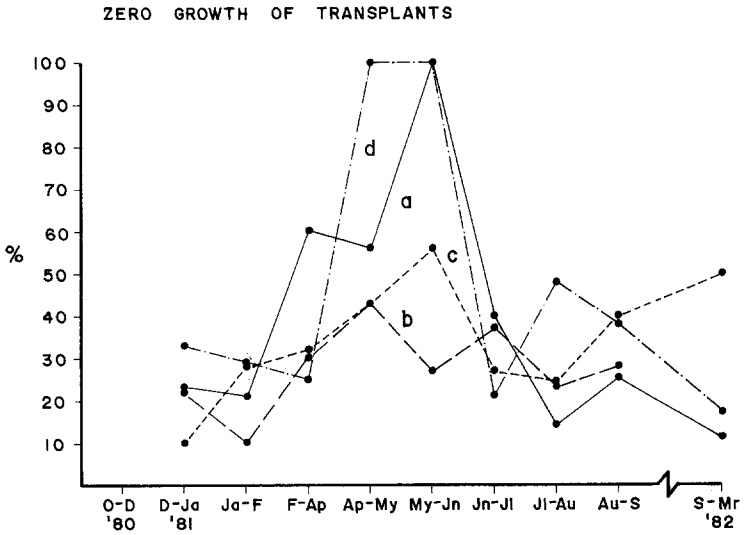

Fig. 3. *Acropora pulchra.* Percentages of branches of transplants that registered zero growth increments for each monthly interval from December 1980 to September 1981, and for the period September 1981–March 1982. (a) Site A transplants; (b) Site B transplants; (c) Module I, Site C transplants; (d) Module IV, Site C transplants

Table 1. *Acropora pulchra.* Regeneration and growth of broken branches

Branch no.	Length of new growth (cm)	Growth period (d)	Projected growth rate (cm yr^{-1})
1	2.3	336	2.5
2	13.9	336	15.1
3	8.2	336	8.9
4	14.7	336	16.0
5	20.9	336	22.7
6	21.1	336	22.9
7	13.4	336	14.6
8	6.4	302	7.7
9	2.5	302	3.0
Average ± SD:			12.6 ± 7.6

Correlation of trends in day length and temperature with growth, mortality and budding

An assumption in this study is that day length and temperature each varied linearly with growth, mortality and budding. Based on this assumption, simple correlation coefficients (Pearson's Product-Moment Correlation Coefficient, Sokal and Rohlf, 1969) were used to infer relationships between growth, mortality and budding and each of the two environmental variables (Table 3). Only the biological parameters that showed significant variations over time, viz. growth rates of controls and transplants, percentage zero growth of transplants, percentage mortalities of transplants, and budding of controls were tested.

Growth

In the case of the controls, up to 90% of the variation in growth may be attributed to negative effects of temperature or day length, or both. As can be seen from Table 3, both factors may be considered significant in terms of effects on growth rate, except for controls at Site D in which case temperature appears to be the more dominant variable.

Day length and temperature are negatively correlated in varying degrees with growth rates of the transplants. Temperature appears to be the more significant variable.

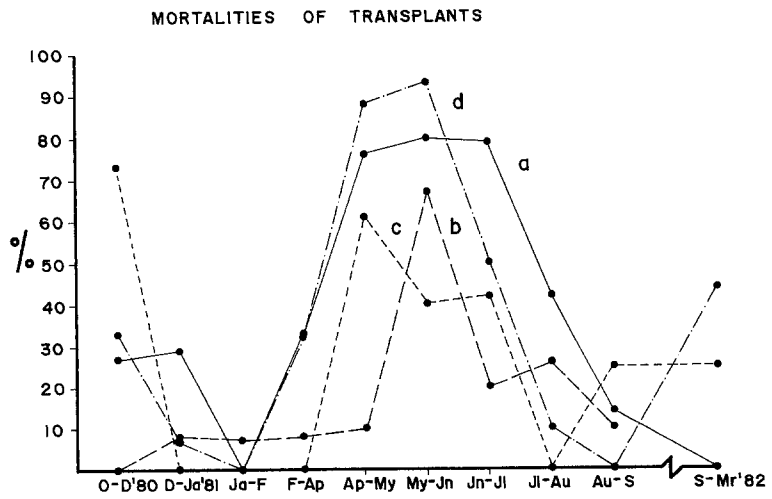

Fig. 4. *Acropora pulchra.* Percentage mortalities of branches of transplants for each monthly interval from October 1980 to September 1981, and for the period September 1981–March 1982. (a) Site A transplants; (b) Site B transplants; (c) Module I, Site C transplants; (d) Module IV, Site C transplants

Table 2. *Acropora pulchra.* Average number of new buds per site for each time interval (n = number of branches observed at the beginning of the study; losses at Sites A, B and D as the study progressed were 30.8, 16 and 6.7% respectively)

Site	Oct – Dec 1980	Dec 1980 – Jan 1981	Jan – Feb 1981	Feb – Apr 1981	Apr – May 1981	May – June 1981	June – July 1981	July – Aug 1981	Aug – Sept 1981
Site A (26)	1.96	1.77	1.28	0.36	0.53	0.22	0.06	0.42	0.53
Site B (25)	1.53	1.42	0.83	0.29	0.27	0.10	0.15	0.58	0.60
Site D (30)		0.50	1.10	0.20	0.21	0.04	0.86	0.68	0.89

Percentages of branches of transplants at all sites exhibiting zero growth are significantly positively correlated with temperature alone. Consistent positive correlations, though not significant, also exist with respect to day length.

Mortality

No attempt was made to correlate day length with percentage mortality of transplants as this factor is normally not known to bring about coral mortality. A positive correlation exists between percentage mortality and temperature, and is significant for transplants at Sites A and C.

Table 3. *Acropora pulchra.* Correlation coefficients (r) showing the relationship between certain biological parameters and the environmental factors, day length and temperature (** significant at the 5% level; * significant at the 10% level)

Site	Day length	Temperature
A. Growth rates of controls		
A	−0.88157**	−0.92749**
B	−0.88819**	−0.92208**
D	−0.57205	−0.91009**
B. Growth rates of transplants		
A	−0.40057	−0.70589
B	−0.70350*	−0.81044*
C, I	−0.78186**	−0.80455*
C, IV	−0.51576	−0.82585**
C. Percentage zero growth of transplants		
A	0.44337	0.79823*
B	0.60410	0.79912*
C, I	0.59034	0.76407*
C, IV	0.41967	0.74684*
D. Percentage mortality of transplants		
A		0.95961**
B		0.63204
C, I		0.88727**
C, IV		0.98107**
E. Budding of controls		
A	−0.9149261**	−0.6219097
B	−0.8677062**	−0.6134047
D	−0.2268210	−0.6523813

Budding

In control colonies at Sites A and B, budding is significantly negatively correlated with day length. Budding of corals at Site D, on the other hand, does not correlate with day length. Temperature in all cases appears to exert no significant effect.

Discussion

Seasonality in coral growth and branching

Based on the results of this study, coral growth within one year is perceived to exhibit a certain pattern. High growth rates occur during the relatively cooler times of the year, and low growth rates during the warmer periods. Cool months in the Philippines coincide roughly with the greatest intensity of the northeast monsoon, which is approximately from November to January. During the transition to the other monsoon regime in the country, the southwest, temperatures observed during the period of the study were rather variable, but increased on the average. Temperatures then decreased gradually during the rainy months of July through October towards the shift once more to the northeast monsoon. It is possible that the apparent seasonality in coral growth observed in Bolinao is a more general phenomenon, particularly with respect to temperate species which are subject to larger temperature fluctuations. Knowledge of seasonality would have applications in management, such as in coral transplantation and harvesting schemes.

Preliminary observations indicate that the appearance of new branches likewise exhibits a certain seasonality, being more marked during the cooler months of October to February. Shinn (1976) observed that staghorn coral consistently initiated branching between December and February. In this study, there appears to be a connection between high growth rates and prolificity of branching,

and the two may have been influenced by a common factor or set of factors.

Projected annual growth rate

Using data for net growth (the value at the beginning subtracted from the value at the end) measured over the entire observation period from October 1980 to March 1982, and averaging for the controls at all sites, a value of 15.3 cm yr^{-1} was derived (Table 4, I). In the hope of obtaining more representative values for a year's growth, the above 17-month period was split into two slightly overlapping annual cycles, October 1980 to September 1981, and February 1981 to March 1982. Mean growth rates thus computed were 14.8 and 13.8 cm yr^{-1}, respectively (Tables 4, II and III). Statistical tests (ANOVA, 0.95 confidence coefficient), however, indicated no significant differences among the values yielded by these three different treatments.

If mean monthly growth rates are used as a basis for computation, figures ranging from 13.1 to 15.8 cm yr^{-1} are obtained. Broken branches, on the other hand, were observed to regenerate an average new growth of 12.6 cm within a year's time. All these values are close to the above estimates, and agree well with those quoted in the literature. Mayor (1924) gave an estimated annual growth rate of 12.0 cm yr^{-1} for *Acropora pulchra* and related forms in Samoa. This figure appears to conform to that of 10.0 to 12.5 cm yr^{-1} given by Guppy (1889) and Wood-Jones (1912) for *Acropora* spp. at Cocos-Keeling Atoll in the Indian Ocean (both cited in Mayor, 1924).

Likewise, Tamura and Hada (1932) discovered *Acropora pulchra* to be the most rapid grower among the coral species covered by their study. Their observations yielded a projected annual growth rate of 22.58 cm yr^{-1} for this species at Yap Island in the Carolines, at depths ranging from 0.5 to 2 m. This is higher than the result of this study, probably because their figure was derived from a monitoring period of only 58 d, comprising the relatively cool months of October to December. Our earlier paper (Yap and Gomez, 1981) also quoted a slightly higher figure, 18.1 cm yr^{-1}, again because this represented a less complete cycle, leaving out some of the warm months. If the same monitoring period as that of Tamura and Hada (1932), i.e. October to December, were applied to the study of the species in Bolinao, a projected annual growth rate of 20.9 cm yr^{-1} would be obtained. This is very close to their estimate of 22.58 cm yr^{-1}.

It would therefore seem that period or duration of observation tends to affect estimates of growth rate significantly. This constitutes an artifact to be avoided as much as possible in field investigations.

Possible effects of environmental factors

The parameter day length, which is the total number of hours of daylight, is considered here to represent light, although it is only one of several ways of representing or measuring this environmental factor. It gives an indication, at the very least, of the quantity of light that was available, or that the corals were exposed to. The times of the year in the Philippines characterized by longer day lengths, the summer months, were also associated with clear skies and, thus, more intense solar insolation. It is to be expected that the extended periods of high light intensities related to the longer day lengths and clear skies of summer resulted in warmer ambient temperatures, particularly in the shallow depths characteristic of the habitat of *Acropora pulchra*.

In this study, growth rate was negatively correlated with both day length and temperature. This trend was highly significant in the case of the controls, and was

Table 4. *Acropora pulchra.* Projected annual growth rates (values stand for mean ± SD cm yr^{-1})

	Site A controls	Site B controls	Site D controls	Mean
I. Observation period: October 1980–March 1982				
Based on net growth over entire observation period	16.8±6.3	15.2±9.7	13.9±9.6	15.3
Based on mean monthly growth rates	16.1	16.1	15.2	15.8
II. Observation period: October 1980–September 1981				
Based on net growth over entire observation period	15.8±5.0	13.8±8.6	14.8±8.5	14.8
Based on mean monthly growth rates	16.0	15.7	15.3	15.7
III. Observation period: February 1981–March 1982				
Based on net growth over entire observation period	14.3±6.9	13.8±10.7	13.3±9.7	13.8
Based on mean monthly growth rates	12.3	13.4	13.7	13.1

discernible in the case of the transplants. The warm summer months may then be considered a probable stressful period for growth of the species.

Large populations of tropical species (and certain subtropical species) live at temperatures precariously close to their upper lethal limits during the summer (Coles *et al.*, 1976). Warming of the water, therefore, from the cool to the warm season implies approaching these upper limits for growth and survival, so that lowered growth rates may be expected (see also discussion in Yap and Gomez, 1981).

The lowest mean temperature recorded at Bolinao, 26 °C, was also the temperature associated with the highest mean growth rates. It is unknown whether growth would have continued to rise had mean temperatures dropped below this level, or if this point already constituted an optimum. Optimum temperatures of 26 °C for *Pocillopora damicornis, Montipora verrucosa* and *Fungia scutaria,* and 27 °C for *P. damicornis* were postulated by Jokiel and Coles (1977) and Clausen (1971), respectively. Growth rates in both studies were shown to decline above (as well as below) these values, growth behaving in a curvilinear fashion with temperature. A similar trend may apply to *Acropora pulchra.*

In addition, light and temperature may have acted synergistically in terms of their influence on coral growth. That light and temperature, among other environmental variables, can behave synergistically, is indicated by the observations of Coles and Jokiel (1978). In their study, the effect of light was most clearly evident at near-lethal temperatures. High light intensities at these temperatures were shown to interfere with vital temperature-related physiological mechanisms of the coral, such as calcification. This again could explain the low growth rates of *Acropora pulchra* during the warm months, when possible stress to the corals was seen in paler colors attributable to "bleaching" or expulsion of the algal symbionts – an observation also made by Buskirk *et al.* (1981) on *Goniastrea retiformis* in Vanuatu.

In the case of the appearance of new branches, it is not clear which environmental factors are operative. There are indications, though, that reduced daylight hours, more than reduced temperatures, may play a role in the initiation of budding. Other factors, however, should be investigated as well.

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